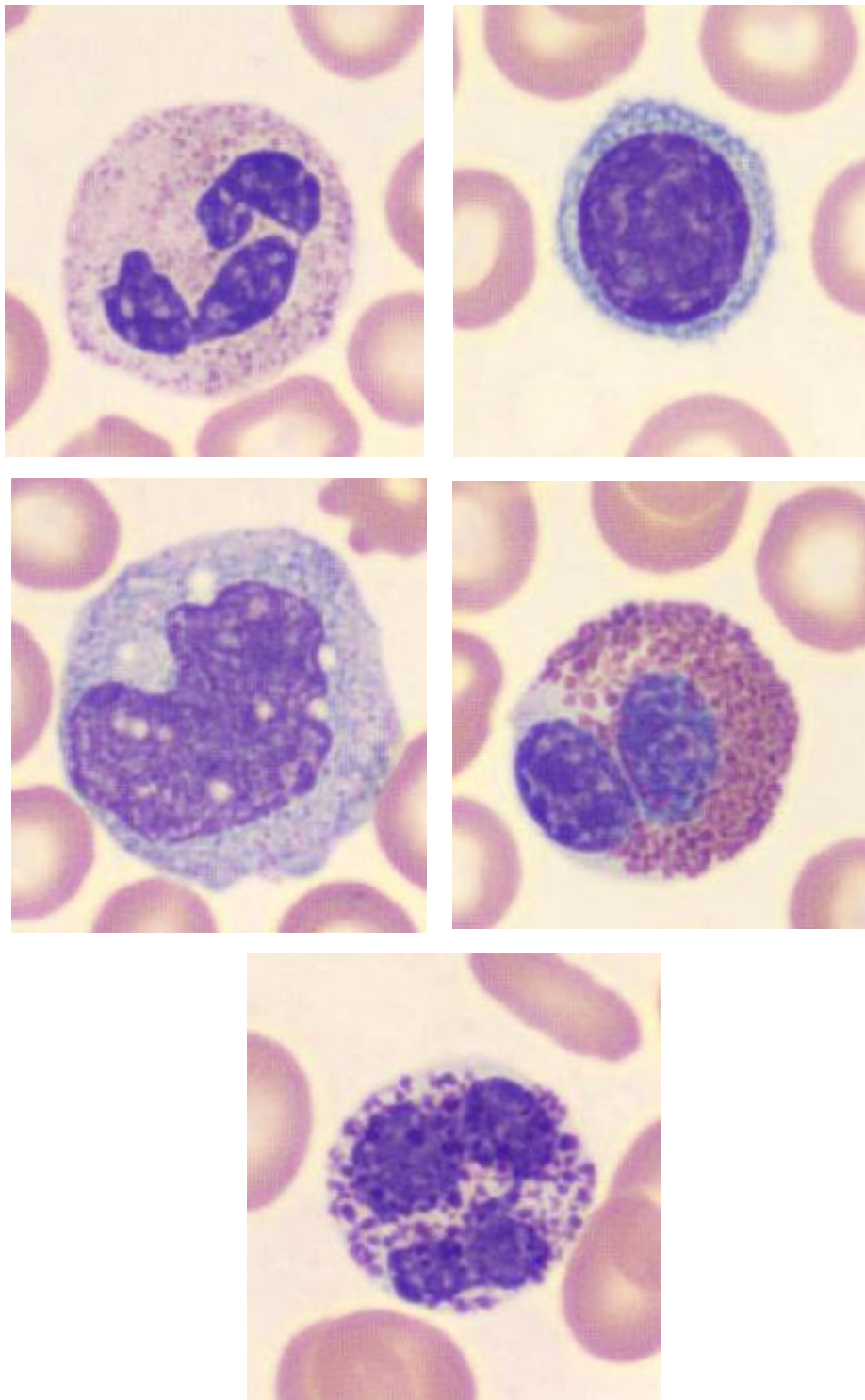


Leukocytes



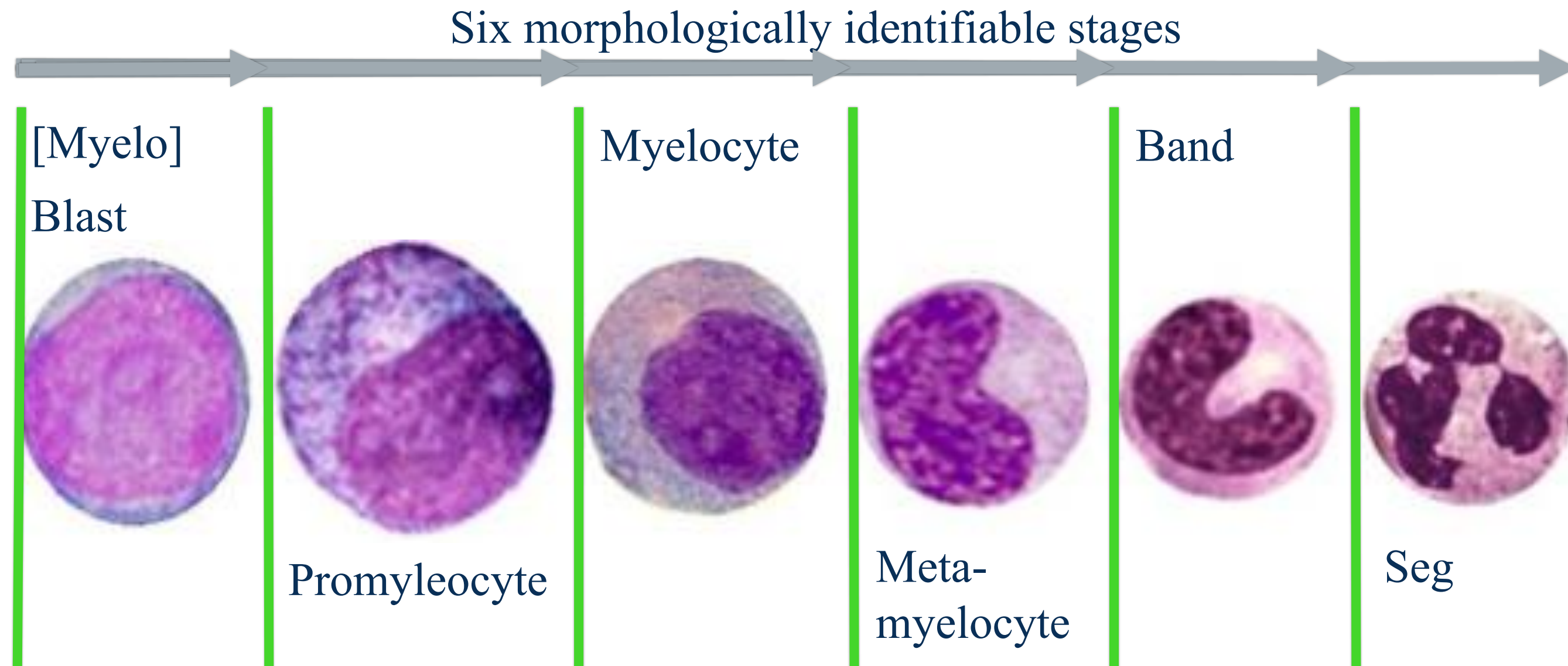
Mature Leukocytes

- Concentrations (CBCD) affected by
 - Age, physiologic and pathologic events
- Fight infection
 - Phagocytosis
 - Neutrophils (40-80%) + Monocytes (2-10%)
 - Adaptive immune response
 - Lymphocytes (25-35%) + Monocytes (2-10%)
 - Eosinophils (0-5%), Basophils (0-1%)

Leukocyte Function

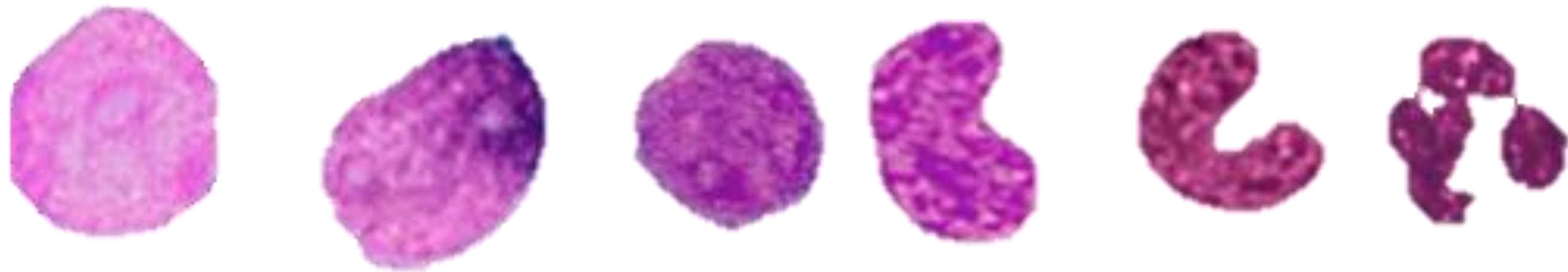
- Briefly circulate in PB before moving into tissues
- Primary functions – protect host from infections
 - Innate (natural) immune response to molecular patterns
 - PAMPs = pathogen-associated
 - DAMPs = damage-associated
 - Adaptive (acquired)
 - Lymphoids and lymphoid tissue
 - Slower, more specific, long-lasting

Neutrophils



Neutrophil Maturation

- Nucleus
 - Nucleoli disappear, chromatin condenses towards segmented lobes

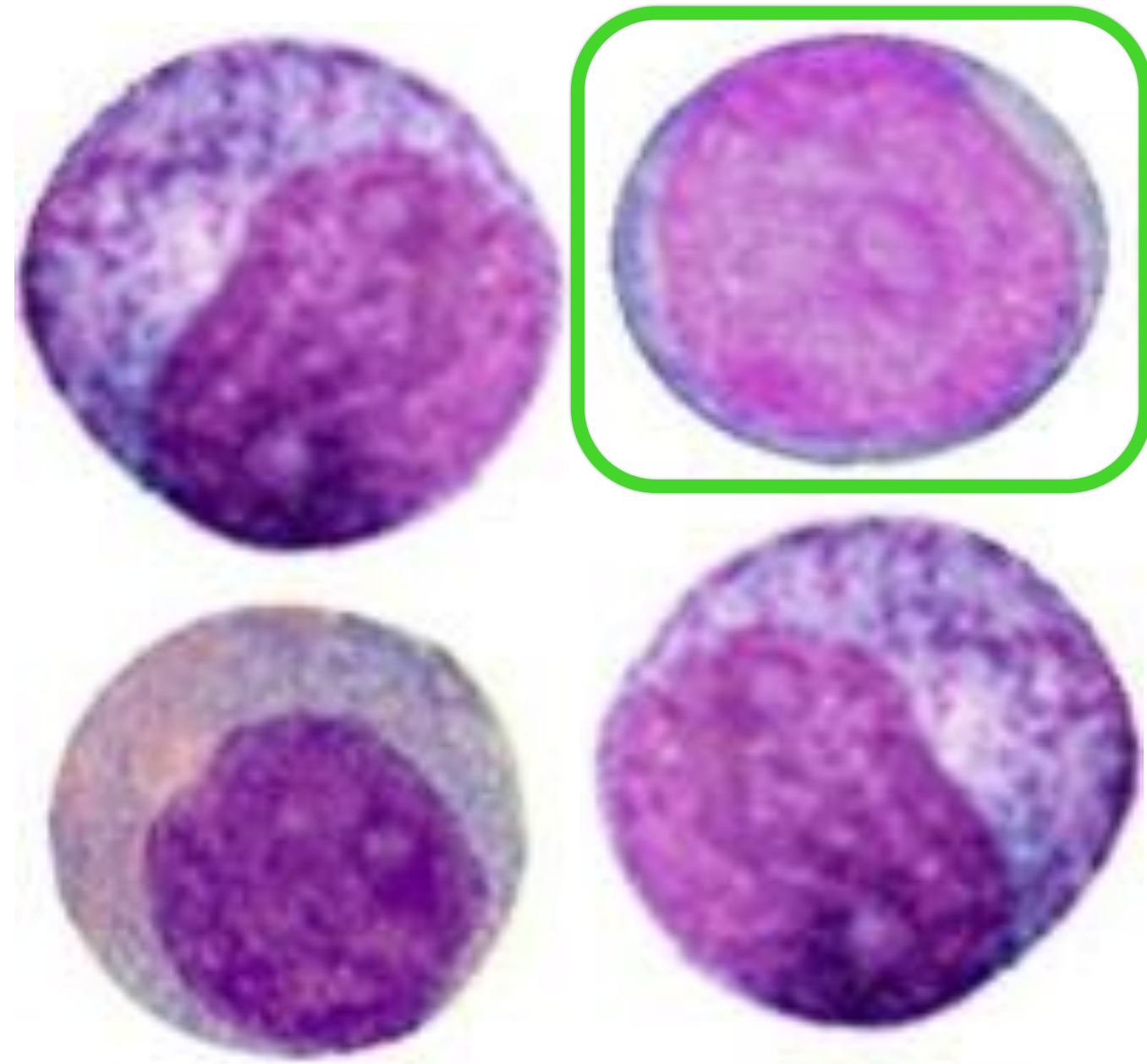


Neutrophil Maturation

- Cytoplasm
 - Scant, agranular, basophilic to pinkish gray, granular
 - Primary (azurophilic) nonspecific granules
 - Secondary specific granules

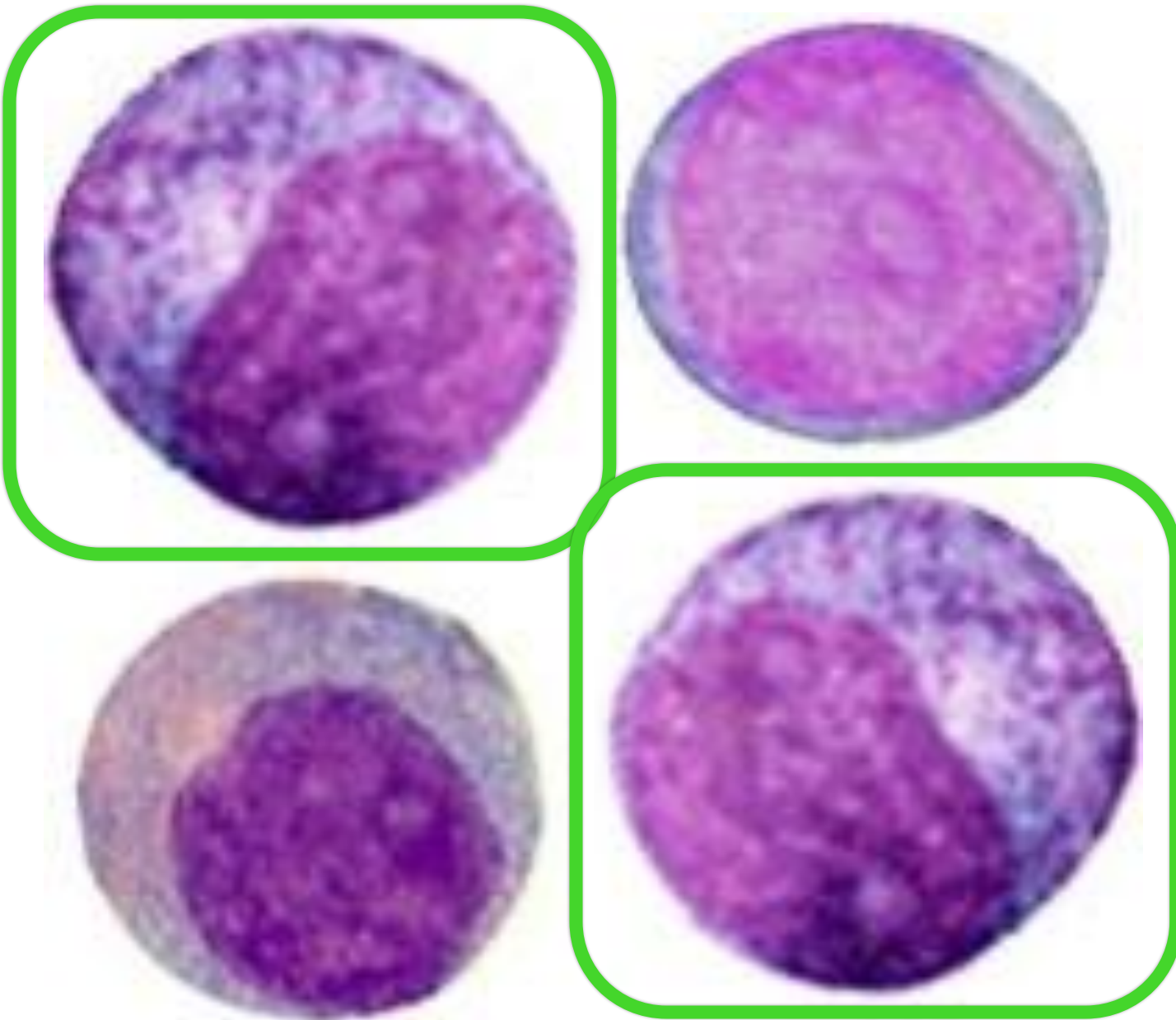


Blast (Myeloblast)



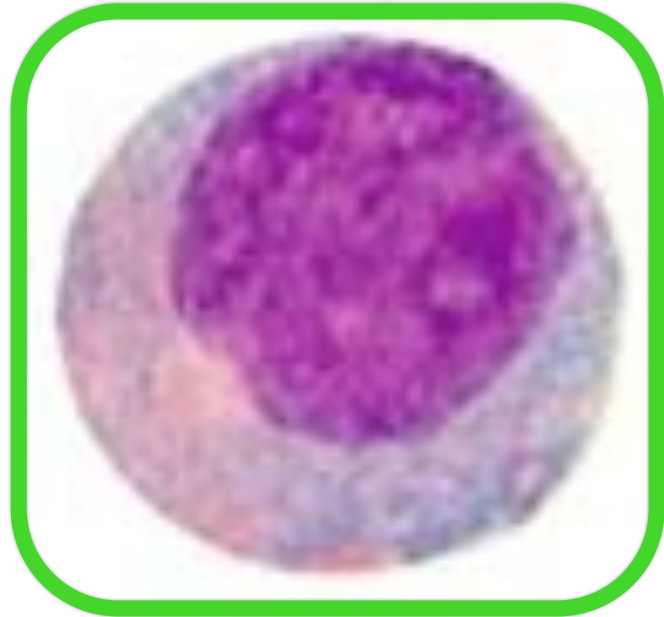
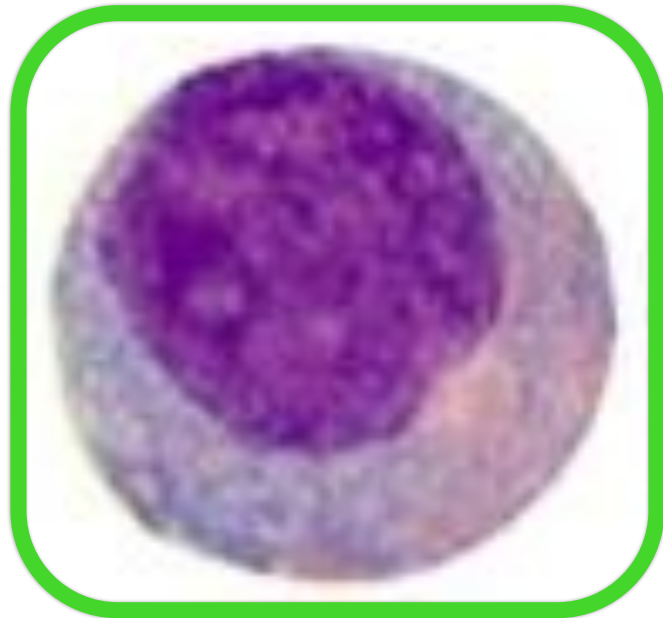
- 14-20 μm , $\uparrow\uparrow\uparrow$ N:C
- Nucleus
 - Round/oval, lacy chromatin, 1-5 nucleoli
- Cytoplasm
 - Scant, blue, agranular
 - Distinct Golgi apparatus
 - < 20 Primary Granules
 - Faint MPO+
 - [CD45, 33, 34, 13]+

Promyelocyte



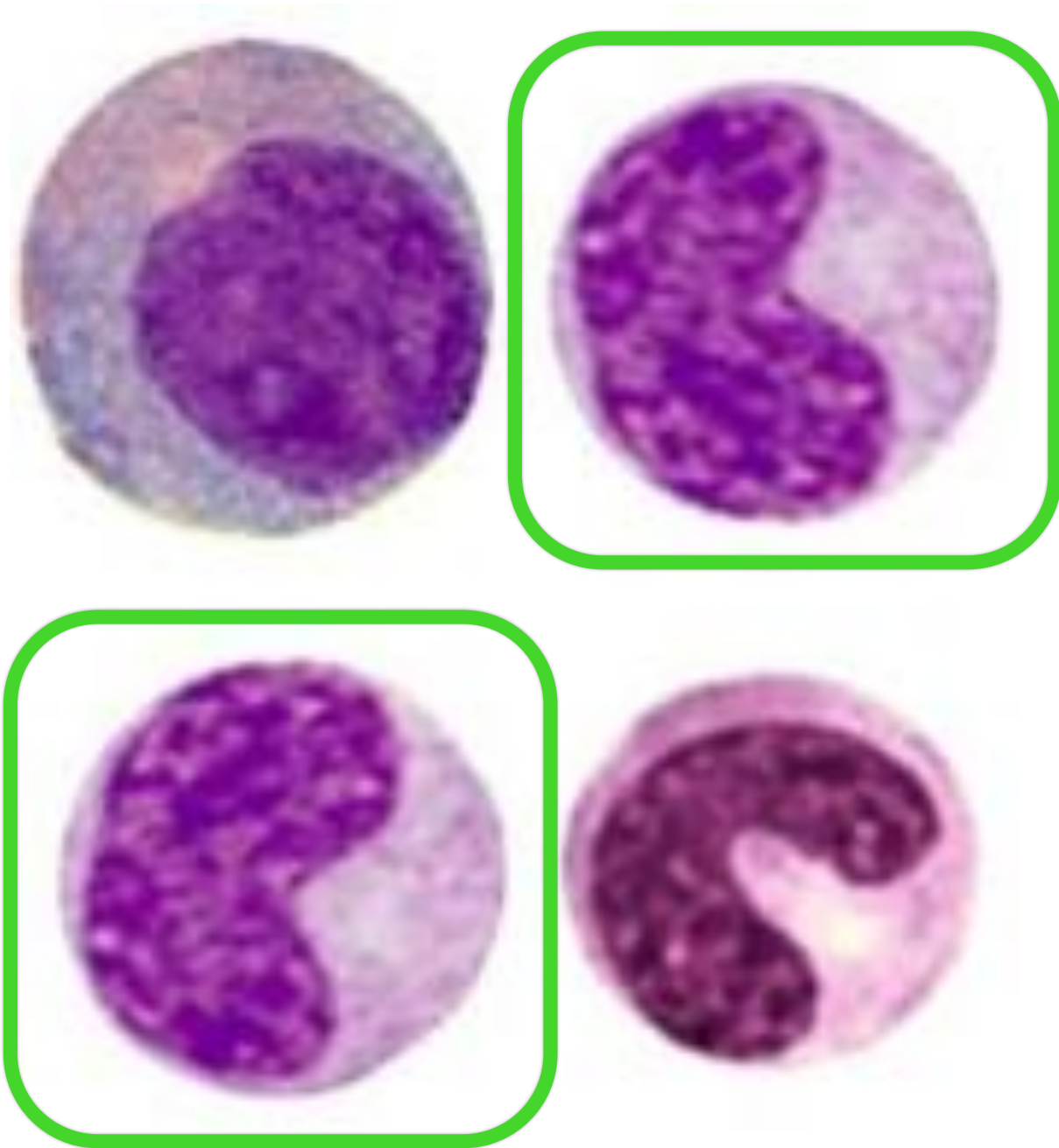
- 15-21 μm , $\uparrow\uparrow$ N:C
- Nucleus
 - Round/oval, chromatin coarse, nucleoli present
- Cytoplasm
 - Basophilic
 - 20+ Primary granules
 - MPO++
 - [CD45, 33, 13]+

Myelocyte



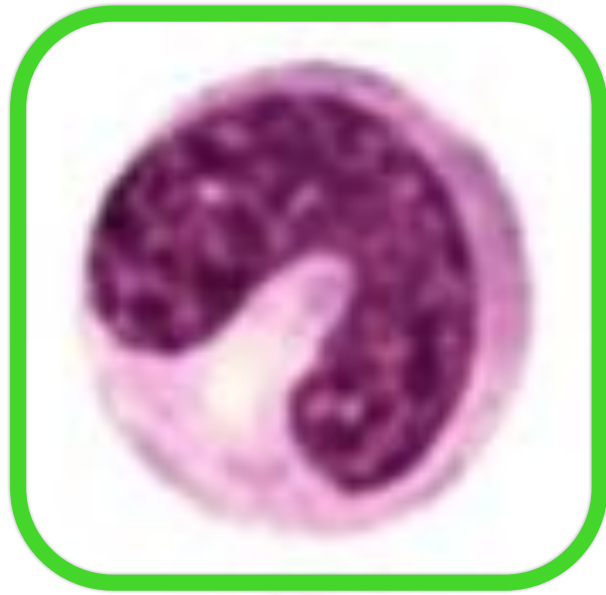
- 12-18 μm , $\uparrow$ N:C
- Nucleus
 - Round/oval/flat, chromatin coarser, nucleoli not apparent
- Cytoplasm
 - Basophilic
 - First to show secondary granules (pink)
 - MPO-
 - Last stage for mitosis

Metamyelocyte



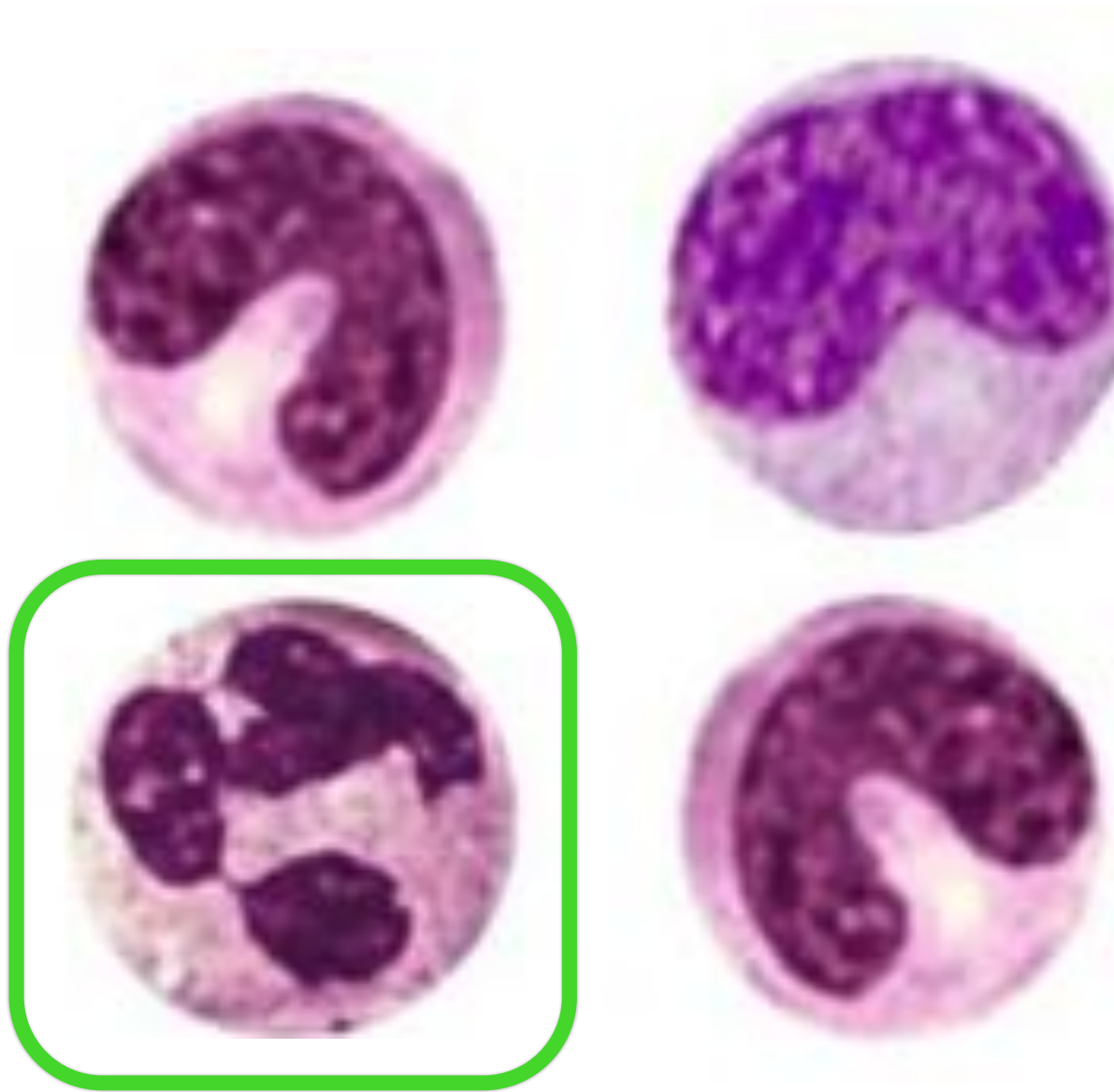
- 10-18 μm
- Nucleus
 - Indented (kidney bean), chromatin coarser/clumped
 - Less than $\frac{1}{2}$ the diameter
- Cytoplasm
 - Neutral pink
 - Secondary granules more abundant

Band Neutrophil

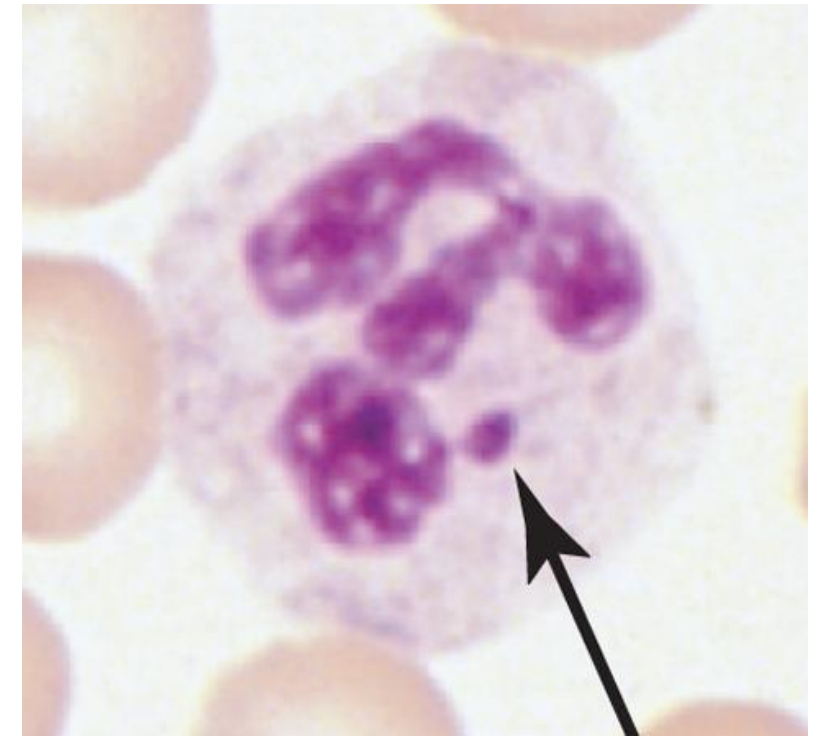


- 9-15 μm
- Nucleus
 - Indent more than $\frac{1}{2}$ the diameter
 - $\uparrow\uparrow$ chromatin condensation at ends
- Cytoplasm
 - Pink-tan
 - Secondary granules

Segmented Neutrophil

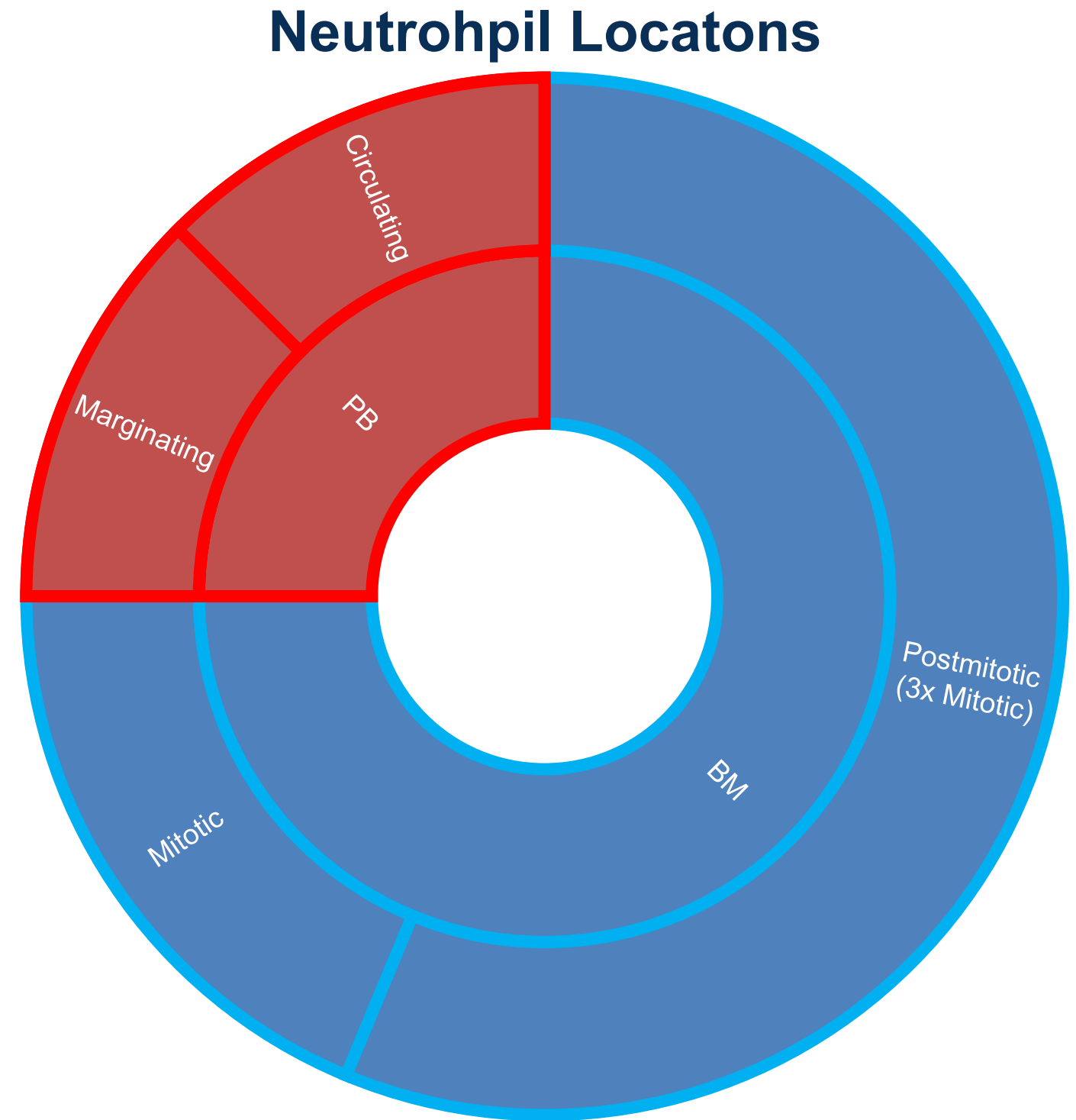


- 9-15 μm
- Nucleus
 - Segmented lobes (3 - 5)
 - Hypo- vs hyper- segmented
- Cytoplasm
 - Pink-tan
 - Secondary + secretory vesicles



Neutrophil Pools

- Life: BM $\rightarrow$ PB $\rightarrow$ Tissues
- Mature neutrophil reserve in BM
 - Steady state maintained from BM to PB
- Two PB pools in equilibrium
 - Circulating [1/2]
 - Marginating [1/2]
 - ~8 hour life span $\rightarrow$ diapedese to tissue



Neutrophil Kinetics

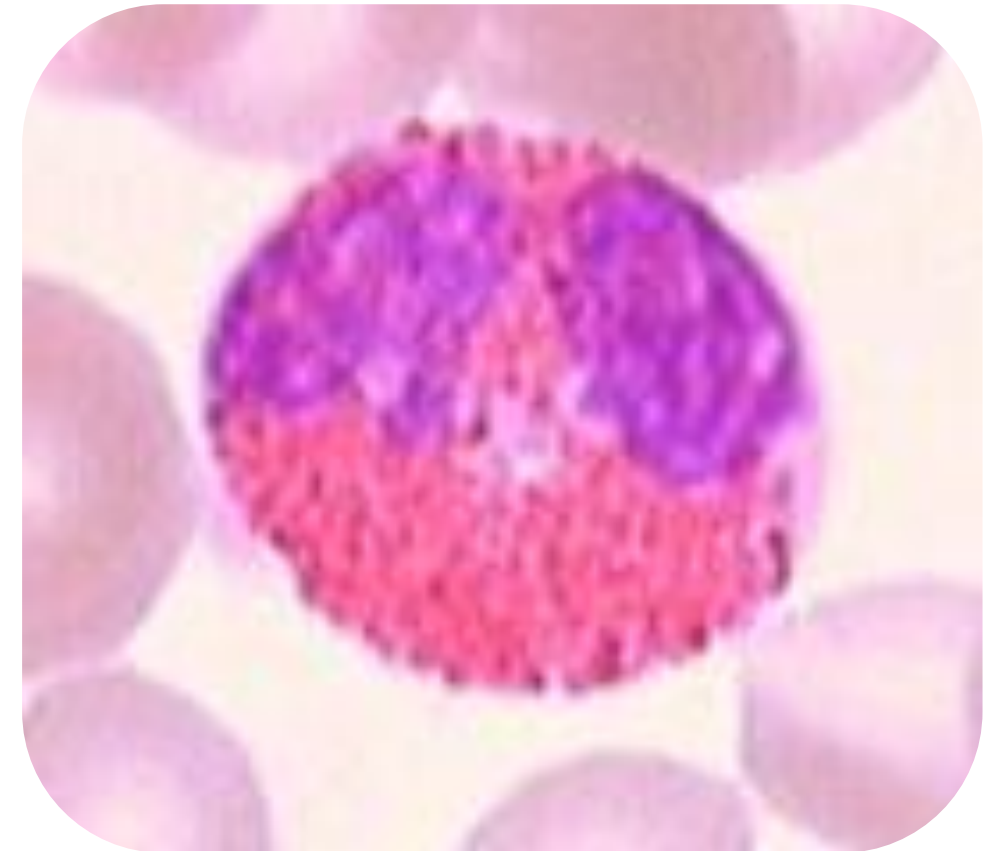
- Signs of a pathological process
 - Neutrophilia [$> 7.0 \times 10^3/\mu\text{L}$]
 - Infection, intoxication, tissue necrosis
 - Granulocytosis
 - Neutropenia [$< 1.8 \times 10^9/\text{L}$]
 - Granulocytopenia [$< 2.0 \times 10^9/\text{L}$]
 - Agranulocytosis

Neutrophil Function

- Innate immune response
 - Adherence > Migration (chemotaxis) > Phagocytosis > Murder bacteria
 - Bacteria Killing
 - Oxygen-dependent/oxidative microbicidal
 - MPO

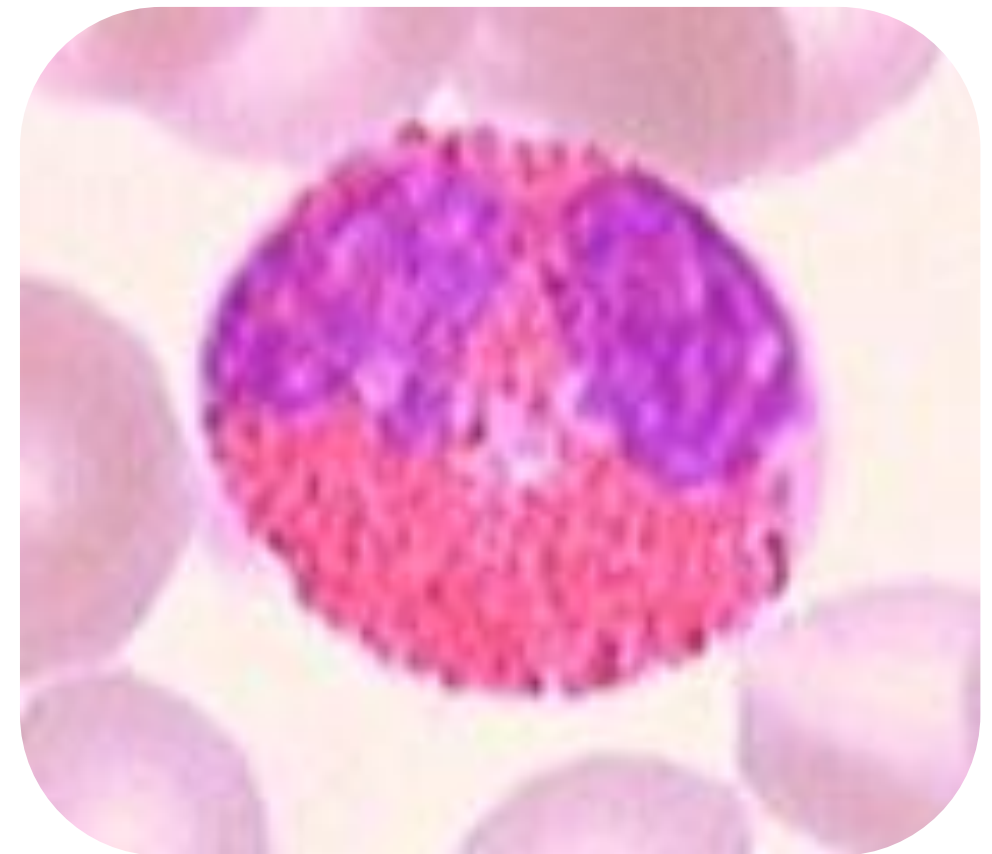
Eosinophils

- Similar maturation stages to neutrophil
 - 1st recognizable stage, eosinophilic myelocyte
- 12-15 μm
- 2-3 lobed nucleus
- Large, eosin-staining, reddish-orange granules
 - Charcot-Leyden proteins



Eosinophil Kinetics

- Diurnal (highest in morning)
- ~18 hours in PB → tissues
- Eosinophilia ($>0.4 \times 10^3/\mu\text{L}$)
 - Parasitic infections



Basophils

- Similar maturation stages to neutrophil
 - 1st recognizable stage, basophilic myelocyte
- 10-15 μm
- 2-3 lobed nucleus
- Large, dark purple/black granules fill cell
 - Histamine, heparin, others
- Immediate hypersensitivity reactions
 - Asthma, urticaria, allergies

